

Thermodynamics

Instructor

Research Assistant Buğra Kaplan

Chapter 6 – Problems

**Problem numbers correspond to
5th Edition**

6-17 A 600-MW steam power plant, which is cooled by a nearby river, has a thermal efficiency of 40 percent. Determine the rate of heat transfer to the river water. Will the actual heat transfer rate be higher or lower than this value? Why?

6–20 A steam power plant with a power output of 150 MW consumes coal at a rate of 60 tons/h. If the heating value of the coal is 30,000 kJ/kg, determine the overall efficiency of this plant. *Answer: 30.0 percent*

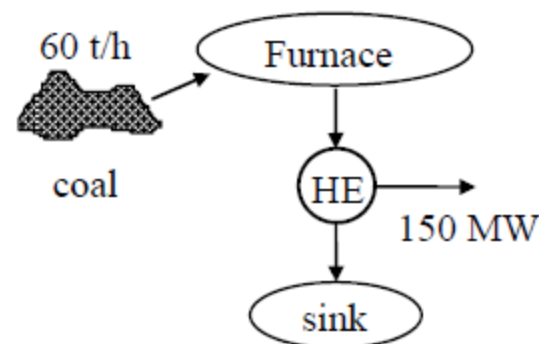
Properties The heating value of coal is given to be 30,000 kJ/kg.

Analysis The rate of heat supply to this power plant is

$$\begin{aligned}\dot{Q}_H &= \dot{m}_{\text{coal}} q_{\text{HV, coal}} \\ &= (60,000 \text{ kg/h})(30,000 \text{ kJ/kg}) = 1.8 \times 10^9 \text{ kJ/h} \\ &= 500 \text{ MW}\end{aligned}$$

Then the thermal efficiency of the plant becomes

$$\eta_{\text{th}} = \frac{\dot{W}_{\text{net, out}}}{\dot{Q}_H} = \frac{150 \text{ MW}}{500 \text{ MW}} = 0.300 = \mathbf{30.0\%}$$



6-21 An automobile engine consumes fuel at a rate of 28 L/h and delivers 60 kW of power to the wheels. If the fuel has a heating value of 44,000 kJ/kg and a density of 0.8 g/cm³, determine the efficiency of this engine. *Answer: 21.9 percent*

6-23 In 2001, the United States produced 51 percent of its electricity in the amount of 1.878×10^{12} kWh from coal-fired power plants. Taking the average thermal efficiency to be 34 percent, determine the amount of thermal energy rejected by the coal-fired power plants in the United States that year.

6-24 The Department of Energy projects that between the years 1995 and 2010, the United States will need to build new power plants to generate an additional 150,000 MW of electricity to meet the increasing demand for electric power. One possibility is to build coal-fired power plants, which cost \$1300 per kW to construct and have an efficiency of 34 percent. Another possibility is to use the clean-burning Integrated Gasification Combined Cycle (IGCC) plants where the coal is subjected to heat and pressure to gasify it while removing sulfur and particulate matter from it. The gaseous coal is then burned in a gas turbine, and part of the waste heat from the exhaust gases is recovered to generate steam for the steam turbine. Currently the construction of IGCC plants costs about \$1500 per kW, but their efficiency is about 45 percent. The average heating value of the coal is about 28,000,000 kJ per ton (that is, 28,000,000 kJ of heat is released when 1 ton of coal is burned). If the IGCC plant is to recover its cost difference from fuel savings in five years, determine what the price of coal should be in \$ per ton.

6-26 Repeat Prob. 6-24 for a simple payback period of three years instead of five years.

6–28 A coal-burning steam power plant produces a net power of 300 MW with an overall thermal efficiency of 32 percent. The actual gravimetric air–fuel ratio in the furnace is calculated to be 12 kg air/kg fuel. The heating value of the coal is 28,000 kJ/kg. Determine (a) the amount of coal consumed during a 24-hour period and (b) the rate of air flowing through the furnace. *Answers: (a) 2.89×10^6 kg, (b) 402 kg/s*

Refrigerators and Heat Pumps

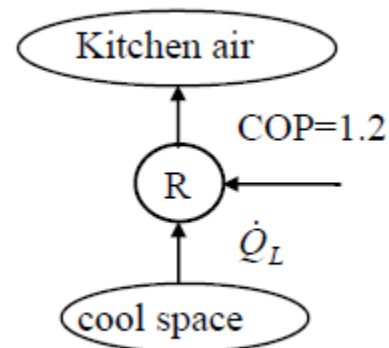
6–39 A household refrigerator with a COP of 1.2 removes heat from the refrigerated space at a rate of 60 kJ/min. Determine (a) the electric power consumed by the refrigerator and (b) the rate of heat transfer to the kitchen air. *Answers: (a) 0.83 kW, (b) 110 kJ/min*

Analysis (a) Using the definition of the coefficient of performance, the power input to the refrigerator is determined to be

$$\dot{W}_{\text{net,in}} = \frac{\dot{Q}_L}{\text{COP}_R} = \frac{60 \text{ kJ/min}}{1.2} = 50 \text{ kJ/min} = \mathbf{0.83 \text{ kW}}$$

(b) The heat transfer rate to the kitchen air is determined from the energy balance,

$$\dot{Q}_H = \dot{Q}_L + \dot{W}_{\text{net,in}} = 60 + 50 = \mathbf{110 \text{ kJ/min}}$$



6-43 A household refrigerator that has a power input of 450 W and a COP of 2.5 is to cool five large watermelons, 10 kg each, to 8°C. If the watermelons are initially at 20°C, determine how long it will take for the refrigerator to cool them. The watermelons can be treated as water whose specific heat is 4.2 kJ/kg · °C. Is your answer realistic or optimistic? Explain. *Answer: 2240 s*

Properties The specific heat of watermelons is given to be $c = 4.2 \text{ kJ/kg} \cdot ^\circ\text{C}$.

Analysis The total amount of heat that needs to be removed from the watermelons is

$$Q_L = (mc\Delta T)_{\text{watermelons}} = 5 \times (10 \text{ kg})(4.2 \text{ kJ/kg} \cdot ^\circ\text{C})(20 - 8)^\circ\text{C} = 2520 \text{ kJ}$$

The rate at which this refrigerator removes heat is

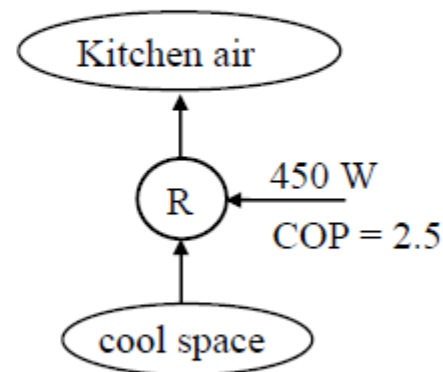
$$\dot{Q}_L = (\text{COP}_R)(\dot{W}_{\text{net,in}}) = (2.5)(0.45 \text{ kW}) = 1.125 \text{ kW}$$

That is, this refrigerator can remove 1.125 kJ of heat per second.

Thus the time required to remove 2520 kJ of heat is

$$\Delta t = \frac{Q_L}{\dot{Q}_L} = \frac{2520 \text{ kJ}}{1.125 \text{ kJ/s}} = 2240 \text{ s} = \mathbf{37.3 \text{ min}}$$

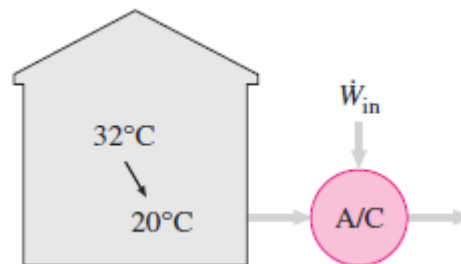
This answer is optimistic since the refrigerated space will gain some heat during this process from the surrounding air, which will increase the work load. Thus, in reality, it will take longer to cool the watermelons.



6-44



When a man returns to his well-sealed house on a summer day, he finds that the house is at 32°C . He turns on the air conditioner, which cools the entire house to 20°C in 15 min. If the COP of the air-conditioning system is 2.5, determine the power drawn by the air conditioner. Assume the entire mass within the house is equivalent to 800 kg of air for which $c_v = 0.72 \text{ kJ/kg} \cdot ^\circ\text{C}$ and $c_p = 1.0 \text{ kJ/kg} \cdot ^\circ\text{C}$.



6-48

A house that was heated by electric resistance heaters consumed 1200 kWh of electric energy in a winter month. If this house were heated instead by a heat pump that has an average COP of 2.4, determine how much money the home owner would have saved that month. Assume a price of 8.5¢/kWh for electricity.

Analysis The amount of heat the resistance heaters supply to the house is equal to the amount of electricity they consume. Therefore, to achieve the same heating effect, the house must be supplied with 1200 kWh of energy. A heat pump that supplied this much heat will consume electrical power in the amount of

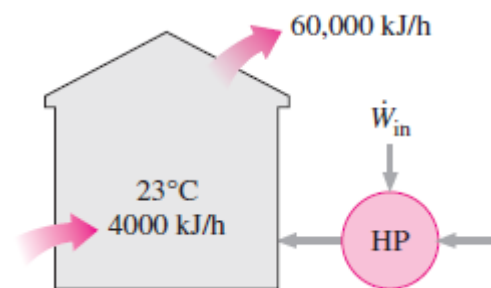
$$\dot{W}_{\text{net,in}} = \frac{\dot{Q}_H}{\text{COP}_{\text{HP}}} = \frac{1200 \text{ kWh}}{2.4} = 500 \text{ kWh}$$

which represent a savings of $1200 - 500 = 700 \text{ kWh}$. Thus the homeowner would have saved

$$(700 \text{ kWh})(0.085 \text{ \$/kWh}) = \text{\$59.50}$$

6-51 A heat pump is used to maintain a house at a constant temperature of 23°C . The house is losing heat to the outside air through the walls and the windows at a rate of $60,000 \text{ kJ/h}$ while the energy generated within the house from people, lights, and appliances amounts to 4000 kJ/h . For a COP of 2.5, determine the required power input to the heat pump.

Answer: 6.22 kW

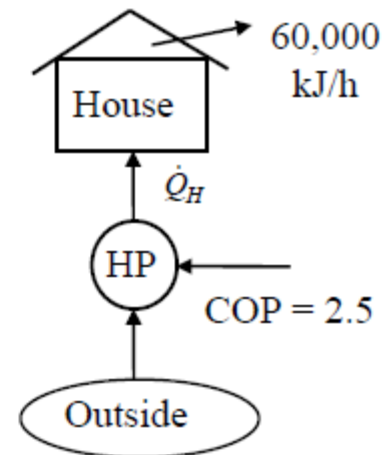


Analysis The heating load of this heat pump system is the difference between the heat lost to the outdoors and the heat generated in the house from the people, lights, and appliances,

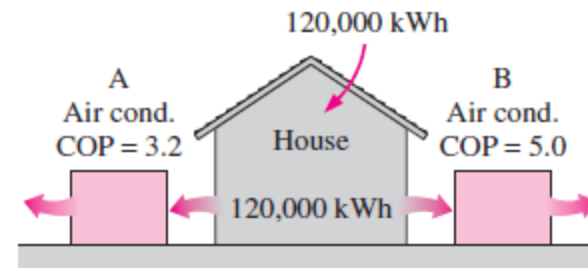
$$\dot{Q}_H = 60,000 - 4,000 = 56,000 \text{ kJ/h}$$

Using the definition of COP, the power input to the heat pump is determined to be

$$\dot{W}_{\text{net,in}} = \frac{\dot{Q}_H}{\text{COP}_{\text{HP}}} = \frac{56,000 \text{ kJ/h}}{2.5} \left(\frac{1 \text{ kW}}{3600 \text{ kJ/h}} \right) = 6.22 \text{ kW}$$



6-53 Consider a building whose annual air-conditioning load is estimated to be 120,000 kWh in an area where the unit cost of electricity is \$0.10/kWh. Two air conditioners are considered for the building. Air conditioner A has a seasonal average COP of 3.2 and costs \$5500 to purchase and install. Air conditioner B has a seasonal average COP of 5.0 and costs \$7000 to purchase and install. All else being equal, determine which air conditioner is a better buy.



6-54 Refrigerant-134a enters the condenser of a residential heat pump at 800 kPa and 35°C at a rate of 0.018 kg/s and leaves at 800 kPa as a saturated liquid. If the compressor consumes 1.2 kW of power, determine (a) the COP of the heat pump and (b) the rate of heat absorption from the outside air.

Assumptions 1 The heat pump operates steadily. 2 The kinetic and potential energy changes are zero.

Properties The enthalpies of R-134a at the condenser inlet and exit are

$$\left. \begin{array}{l} P_1 = 800 \text{ kPa} \\ T_1 = 35^\circ\text{C} \end{array} \right\} h_1 = 271.22 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_2 = 800 \text{ kPa} \\ x_2 = 0 \end{array} \right\} h_2 = 95.47 \text{ kJ/kg}$$

Analysis (a) An energy balance on the condenser gives the heat rejected in the condenser

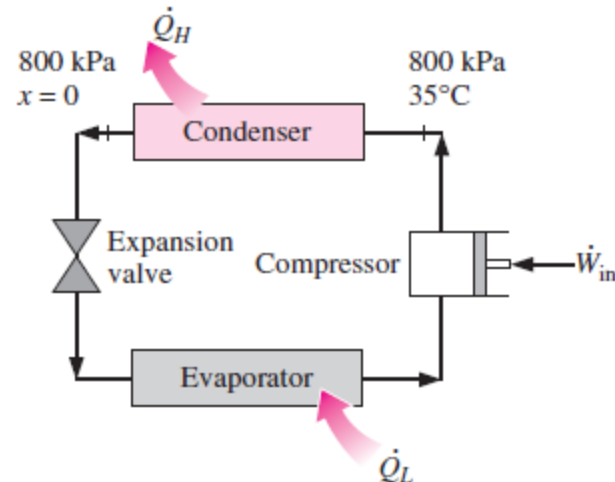
$$\dot{Q}_H = \dot{m}(h_1 - h_2) = (0.018 \text{ kg/s})(271.22 - 95.47) \text{ kJ/kg} = 3.164 \text{ kW}$$

The COP of the heat pump is

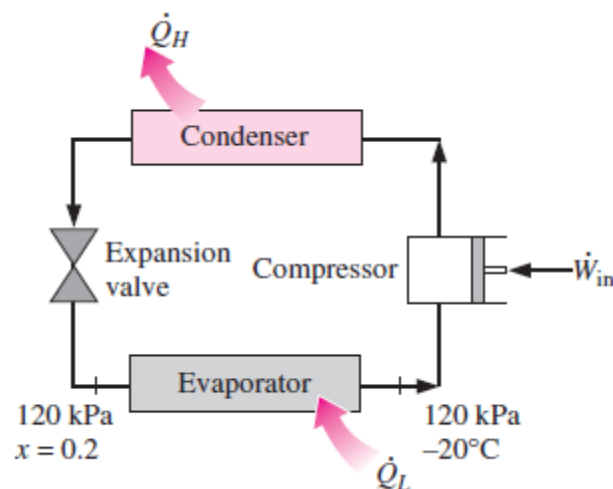
$$\text{COP} = \frac{\dot{Q}_H}{\dot{W}_{\text{in}}} = \frac{3.164 \text{ kW}}{1.2 \text{ kW}} = \mathbf{2.64}$$

(b) The rate of heat absorbed from the outside air

$$\dot{Q}_L = \dot{Q}_H - \dot{W}_{\text{in}} = 3.164 - 1.2 = \mathbf{1.96 \text{ kW}}$$



6–55 Refrigerant-134a enters the evaporator coils placed at the back of the freezer section of a household refrigerator at 120 kPa with a quality of 20 percent and leaves at 120 kPa and -20°C . If the compressor consumes 450 W of power and the COP the refrigerator is 1.2, determine (a) the mass flow rate of the refrigerant and (b) the rate of heat rejected to the kitchen air. *Answers: (a) 0.00311 kg/s, (b) 990 W*



Carnot Heat Engines

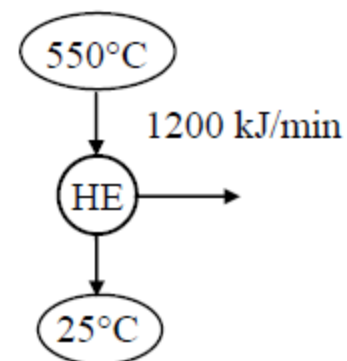
6–73  A heat engine operates between a source at 550°C and a sink at 25°C . If heat is supplied to the heat engine at a steady rate of 1200 kJ/min, determine the maximum power output of this heat engine.

Analysis The highest thermal efficiency a heat engine operating between two specified temperature limits can have is the Carnot efficiency, which is determined from

$$\eta_{\text{th,max}} = \eta_{\text{th,C}} = 1 - \frac{T_L}{T_H} = 1 - \frac{298 \text{ K}}{823 \text{ K}} = 0.638 \text{ or } 63.8\%$$

Then the maximum power output of this heat engine is determined from the definition of thermal efficiency to be

$$\dot{W}_{\text{net,out}} = \eta_{\text{th}} \dot{Q}_H = (0.638)(1200 \text{ kJ/min}) = 765.6 \text{ kJ/min} = \mathbf{12.8 \text{ kW}}$$



6-78 An inventor claims to have developed a heat engine that receives 700 kJ of heat from a source at 500 K and produces 300 kJ of net work while rejecting the waste heat to a sink at 290 K. Is this a reasonable claim? Why?

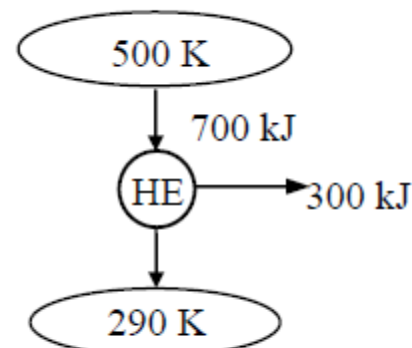
Analysis The highest thermal efficiency a heat engine operating between two specified temperature limits can have is the Carnot efficiency, which is determined from

$$\eta_{th,max} = \eta_{th,C} = 1 - \frac{T_L}{T_H} = 1 - \frac{290 \text{ K}}{500 \text{ K}} = 0.42 \text{ or } 42\%$$

The actual thermal efficiency of the heat engine in question is

$$\eta_{th} = \frac{W_{net}}{Q_H} = \frac{300 \text{ kJ}}{700 \text{ kJ}} = 0.429 \text{ or } 42.9\%$$

which is greater than the maximum possible thermal efficiency. Therefore, this heat engine is a PMM2 and the claim is **false**.



6–80 A geothermal power plant uses geothermal water extracted at 160°C at a rate of 440 kg/s as the heat source and produces 22 MW of net power. If the environment temperature is 25°C, determine (a) the actual thermal efficiency, (b) the maximum possible thermal efficiency, and (c) the actual rate of heat rejection from this power plant.

Properties Using saturated liquid properties, the source and the sink state enthalpies of geothermal water are (Table A-4)

$$\left. \begin{array}{l} T_{\text{source}} = 160^{\circ}\text{C} \\ x_{\text{source}} = 0 \end{array} \right\} h_{\text{source}} = 675.47 \text{ kJ/kg}$$

$$\left. \begin{array}{l} T_{\text{sink}} = 25^{\circ}\text{C} \\ x_{\text{sink}} = 0 \end{array} \right\} h_{\text{sink}} = 104.83 \text{ kJ/kg}$$

Analysis (a) The rate of heat input to the plant may be taken as the enthalpy difference between the source and the sink for the power plant

$$\dot{Q}_{\text{in}} = \dot{m}_{\text{geo}} (h_{\text{source}} - h_{\text{sink}}) = (440 \text{ kg/s})(675.47 - 104.83) \text{ kJ/kg} = 251,083 \text{ kW}$$

The actual thermal efficiency is

$$\eta_{\text{th}} = \frac{\dot{W}_{\text{net,out}}}{\dot{Q}_{\text{in}}} = \frac{22 \text{ MW}}{251.083 \text{ MW}} = \mathbf{0.0876 = 8.8\%}$$

(b) The maximum thermal efficiency is the thermal efficiency of a reversible heat engine operating between the source and sink temperatures

$$\eta_{\text{th,max}} = 1 - \frac{T_L}{T_H} = 1 - \frac{(25 + 273) \text{ K}}{(160 + 273) \text{ K}} = \mathbf{0.312 = 31.2\%}$$

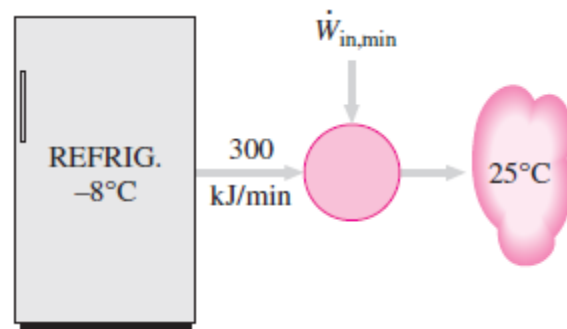
(c) Finally, the rate of heat rejection is

$$\dot{Q}_{\text{out}} = \dot{Q}_{\text{in}} - \dot{W}_{\text{net,out}} = 251.1 - 22 = \mathbf{229.1 \text{ MW}}$$

Carnot Refrigerators and Heat Pumps

6–87 A refrigerator is to remove heat from the cooled space at a rate of 300 kJ/min to maintain its temperature at -8°C . If the air surrounding the refrigerator is at 25°C , determine the minimum power input required for this refrigerator.

Answer: 0.623 kW

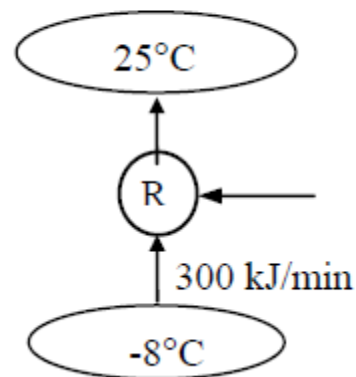


Analysis The power input to a refrigerator will be a minimum when the refrigerator operates in a reversible manner. The coefficient of performance of a reversible refrigerator depends on the temperature limits in the cycle only, and is determined from

$$COP_{R,rev} = \frac{1}{(T_H/T_L) - 1} = \frac{1}{(25 + 273 \text{ K})/(-8 + 273 \text{ K}) - 1} = 8.03$$

The power input to this refrigerator is determined from the definition of the coefficient of performance of a refrigerator,

$$\dot{W}_{net,in,min} = \frac{\dot{Q}_L}{COP_{R,max}} = \frac{300 \text{ kJ/min}}{8.03} = 37.36 \text{ kJ/min} = \mathbf{0.623 \text{ kW}}$$



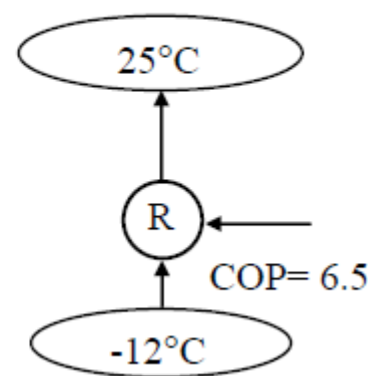
6–90 A Carnot refrigerator operates in a room in which the temperature is 25°C . The refrigerator consumes 500 W of power when operating and has a COP of 4.5. Determine (a) the rate of heat removal from the refrigerated space and (b) the temperature of the refrigerated space. *Answers: (a) 135 kJ/min, (b) -29.2°C*

6-91 An inventor claims to have developed a refrigeration system that removes heat from the closed region at -12°C and transfers it to the surrounding air at 25°C while maintaining a COP of 6.5. Is this claim reasonable? Why?

Analysis The highest coefficient of performance a refrigerator can have when removing heat from a cool medium at -12°C to a warmer medium at 25°C is

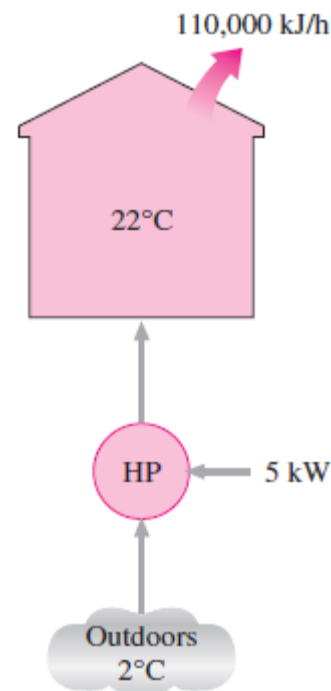
$$\text{COP}_{R,\max} = \text{COP}_{R,\text{rev}} = \frac{1}{(T_H/T_L) - 1} = \frac{1}{(25 + 273 \text{ K})/(-12 + 273 \text{ K}) - 1} = 7.1$$

The COP claimed by the inventor is 6.5, which is below this maximum value, thus the claim is **reasonable**. However, it is not probable.



6-94 A heat pump is used to heat a house and maintain it at 24°C . On a winter day when the outdoor air temperature is -5°C , the house is estimated to lose heat at a rate of 80,000 kJ/h. Determine the minimum power required to operate this heat pump.

6–95 A heat pump is used to maintain a house at 22°C by extracting heat from the outside air on a day when the outside air temperature is 2°C. The house is estimated to lose heat at a rate of 110,000 kJ/h, and the heat pump consumes 5 kW of electric power when running. Is this heat pump powerful enough to do the job?

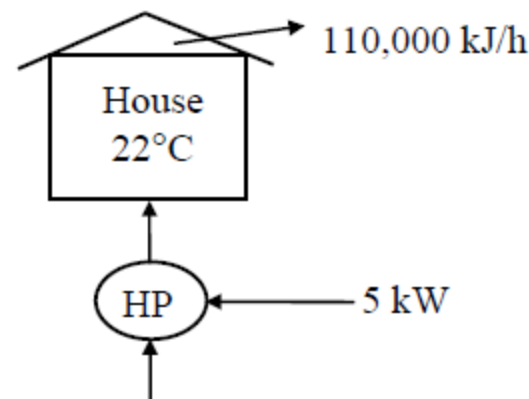


Analysis The power input to a heat pump will be a minimum when the heat pump operates in a reversible manner. The coefficient of performance of a reversible heat pump depends on the temperature limits in the cycle only, and is determined from

$$\text{COP}_{\text{HP,rev}} = \frac{1}{1 - (T_L / T_H)} = \frac{1}{1 - (2 + 273 \text{ K}) / (22 + 273 \text{ K})} = 14.75$$

The required power input to this reversible heat pump is determined from the definition of the coefficient of performance to be

$$\dot{W}_{\text{net,in,min}} = \frac{\dot{Q}_H}{\text{COP}_{\text{HP}}} = \frac{110,000 \text{ kJ/h} \left(\frac{1 \text{ h}}{3600 \text{ s}} \right)}{14.75} = \mathbf{2.07 \text{ kW}}$$



This heat pump is **powerful enough** since 5 kW > 2.07 kW.

6-96 The structure of a house is such that it loses heat at a rate of 5400 kJ/h per °C difference between the indoors and outdoors. A heat pump that requires a power input of 6 kW is used to maintain this house at 21°C. Determine the lowest outdoor temperature for which the heat pump can meet the heating requirements of this house. *Answer: -13.3°C*

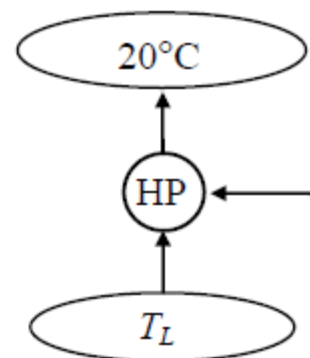
6-97 The performance of a heat pump degrades (i.e., its COP decreases) as the temperature of the heat source decreases. This makes using heat pumps at locations with severe weather conditions unattractive. Consider a house that is heated and maintained at 20°C by a heat pump during the winter. What is the maximum COP for this heat pump if heat is extracted from the outdoor air at (a) 10°C, (b) -5°C, and (c) -30°C?

Analysis The coefficient of performance of a heat pump will be a maximum when the heat pump operates in a reversible manner. The coefficient of performance of a reversible heat pump depends on the temperature limits in the cycle only, and is determined for all three cases above to be

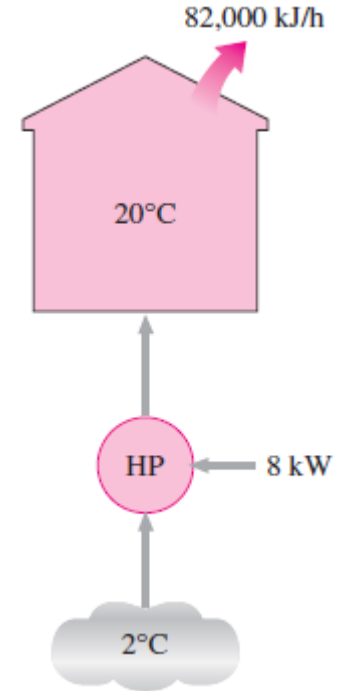
$$\text{COP}_{\text{HP,rev}} = \frac{1}{1 - (T_L / T_H)} = \frac{1}{1 - (10 + 273\text{K}) / (20 + 273\text{K})} = \mathbf{29.3}$$

$$\text{COP}_{\text{HP,rev}} = \frac{1}{1 - (T_L / T_H)} = \frac{1}{1 - (-5 + 273\text{K}) / (20 + 273\text{K})} = \mathbf{11.7}$$

$$\text{COP}_{\text{HP,rev}} = \frac{1}{1 - (T_L / T_H)} = \frac{1}{1 - (-30 + 273\text{K}) / (20 + 273\text{K})} = \mathbf{5.86}$$



6-99 A Carnot heat pump is to be used to heat a house and maintain it at 20°C in winter. On a day when the average outdoor temperature remains at about 2°C , the house is estimated to lose heat at a rate of $82,000\text{ kJ/h}$. If the heat pump consumes 8 kW of power while operating, determine (a) how long the heat pump ran on that day; (b) the total heating costs, assuming an average price of 8.5¢/kWh for electricity; and (c) the heating cost for the same day if resistance heating is used instead of a heat pump. *Answers: (a) 4.19 h, (b) \$2.85, (c) \$46.47*



6-100 A Carnot heat engine receives heat from a reservoir at 900°C at a rate of 800 kJ/min and rejects the waste heat to the ambient air at 27°C . The entire work output of the heat engine is used to drive a refrigerator that removes heat from the refrigerated space at -5°C and transfers it to the same ambient air at 27°C . Determine (a) the maximum rate of heat removal from the refrigerated space and (b) the total rate of heat rejection to the ambient air. *Answers: (a) 4982 kJ/min, (b) 5782 kJ*

Analysis (a) The highest thermal efficiency a heat engine operating between two specified temperature limits can have is the Carnot efficiency, which is determined from

$$\eta_{\text{th,max}} = \eta_{\text{th,C}} = 1 - \frac{T_L}{T_H} = 1 - \frac{300\text{ K}}{1173\text{ K}} = 0.744$$

Then the maximum power output of this heat engine is determined from the definition of thermal efficiency to be

$$\dot{W}_{\text{net,out}} = \eta_{\text{th}} \dot{Q}_H = (0.744)(800\text{ kJ/min}) = 595.2\text{ kJ/min}$$

which is also the power input to the refrigerator, $\dot{W}_{\text{net,in}}$.

The rate of heat removal from the refrigerated space will be a maximum if a Carnot refrigerator is used. The COP of the Carnot refrigerator is

$$\text{COP}_{\text{R,rev}} = \frac{1}{(T_H/T_L) - 1} = \frac{1}{(27 + 273\text{ K})/(-5 + 273\text{ K}) - 1} = 8.37$$

Then the rate of heat removal from the refrigerated space becomes

$$\dot{Q}_{L,R} = (\text{COP}_{\text{R,rev}})(\dot{W}_{\text{net,in}}) = (8.37)(595.2\text{ kJ/min}) = \mathbf{4982\text{ kJ/min}}$$

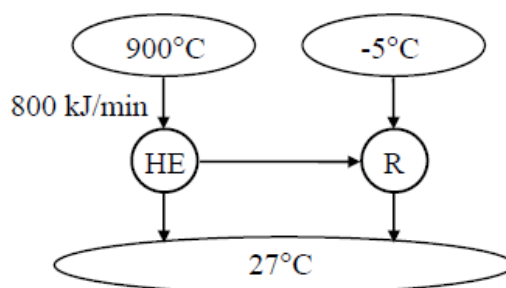
(b) The total rate of heat rejection to the ambient air is the sum of the heat rejected by the heat engine ($\dot{Q}_{L,\text{HE}}$) and the heat discarded by the refrigerator ($\dot{Q}_{H,R}$),

$$\dot{Q}_{L,\text{HE}} = \dot{Q}_{H,\text{HE}} - \dot{W}_{\text{net,out}} = 800 - 595.2 = 204.8\text{ kJ/min}$$

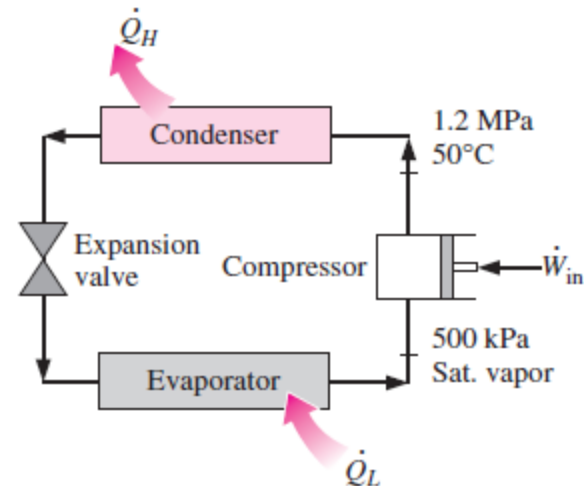
$$\dot{Q}_{H,R} = \dot{Q}_{L,R} + \dot{W}_{\text{net,in}} = 4982 + 595.2 = 5577.2\text{ kJ/min}$$

and

$$\dot{Q}_{\text{ambient}} = \dot{Q}_{L,\text{HE}} + \dot{Q}_{H,R} = 204.8 + 5577.2 = \mathbf{5782\text{ kJ/min}}$$



6-103 An air-conditioner with refrigerant-134a as the working fluid is used to keep a room at 26°C by rejecting the waste heat to the outdoor air at 34°C . The room gains heat through the walls and the windows at a rate of 250 kJ/min while the heat generated by the computer, TV, and lights amounts to 900 W . The refrigerant enters the compressor at 500 kPa as a saturated vapor at a rate of 100 L/min and leaves at 1200 kPa and 50°C . Determine (a) the actual COP, (b) the maximum COP, and (c) the minimum volume flow rate of the refrigerant at the compressor inlet for the same compressor inlet and exit conditions. *Answers: (a) 6.59, (b) 37.4, (c) 17.6 L/min*



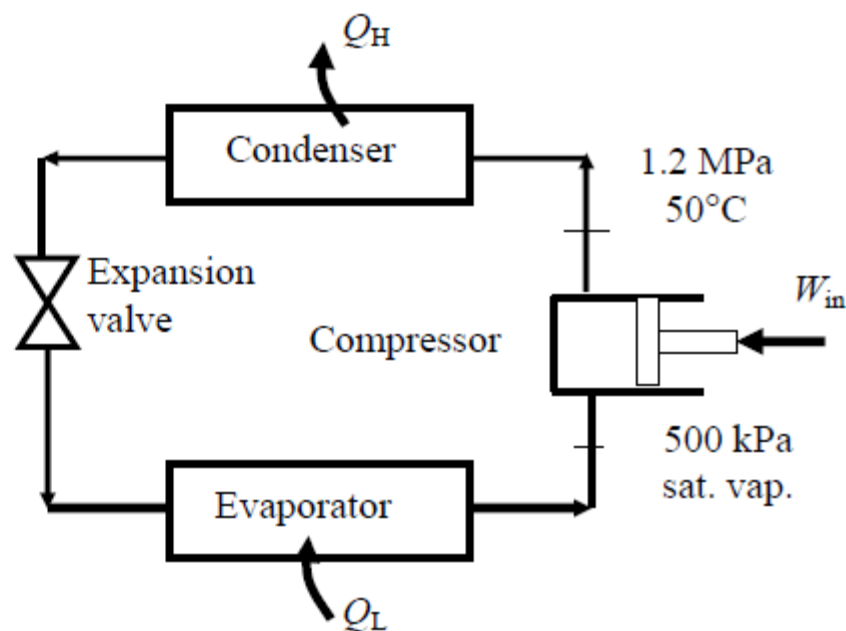
Properties The properties of R-134a at the compressor inlet and exit states are (Tables A-11 through A-13)

$$\left. \begin{array}{l} P_1 = 500\text{ kPa} \\ x_1 = 1 \end{array} \right\} \begin{array}{l} h_1 = 259.30\text{ kJ/kg} \\ v_1 = 0.04112\text{ m}^3/\text{kg} \end{array}$$

$$\left. \begin{array}{l} P_2 = 1.2\text{ MPa} \\ T_2 = 50^\circ\text{C} \end{array} \right\} h_2 = 278.27\text{ kJ/kg}$$

Analysis (a) The mass flow rate of the refrigerant and the power consumption of the compressor are

$$\dot{m}_R = \frac{\dot{V}_1}{v_1} = \frac{100\text{ L/min} \left(\frac{1\text{ m}^3}{1000\text{ L}} \right) \left(\frac{1\text{ min}}{60\text{ s}} \right)}{0.04112\text{ m}^3/\text{kg}} = 0.04053\text{ kg/s}$$



$$\dot{W}_{\text{in}} = \dot{m}_R (h_2 - h_1) = (0.04053 \text{ kg/s})(278.27 - 259.30) \text{ kJ/kg} = 0.7686 \text{ kW}$$

The heat gains to the room must be rejected by the air-conditioner. That is,

$$\dot{Q}_L = \dot{Q}_{\text{heat}} + \dot{Q}_{\text{equipment}} = (250 \text{ kJ/min}) \left(\frac{1 \text{ min}}{60 \text{ s}} \right) + 0.9 \text{ kW} = 5.067 \text{ kW}$$

Then, the actual COP becomes

$$\text{COP} = \frac{\dot{Q}_L}{\dot{W}_{\text{in}}} = \frac{5.067 \text{ kW}}{0.7686 \text{ kW}} = \mathbf{6.59}$$

(b) The COP of a reversible refrigerator operating between the same temperature limits is

$$\text{COP}_{\text{max}} = \frac{1}{T_H / T_L - 1} = \frac{1}{(34 + 273) / (26 + 273) - 1} = \mathbf{37.4}$$

(c) The minimum power input to the compressor for the same refrigeration load would be

$$\dot{W}_{\text{in,min}} = \frac{\dot{Q}_L}{\text{COP}_{\text{max}}} = \frac{5.067 \text{ kW}}{37.38} = 0.1356 \text{ kW}$$

The minimum mass flow rate is

$$\dot{m}_{R,\text{min}} = \frac{\dot{W}_{\text{in,min}}}{h_2 - h_1} = \frac{0.1356 \text{ kW}}{(278.27 - 259.30) \text{ kJ/kg}} = 0.007149 \text{ kg/s}$$

Finally, the minimum volume flow rate at the compressor inlet is

$$\dot{V}_{\text{min},1} = \dot{m}_{R,\text{min}} v_1 = (0.007149 \text{ kg/s})(0.04112 \text{ m}^3/\text{kg}) = 0.000294 \text{ m}^3/\text{s} = \mathbf{17.64 \text{ L/min}}$$

6-113 Consider a Carnot heat-engine cycle executed in a steady-flow system using steam as the working fluid. The cycle has a thermal efficiency of 30 percent, and steam changes from saturated liquid to saturated vapor at 275°C during the heat addition process. If the mass flow rate of the steam is 3 kg/s , determine the net power output of this engine, in kW.

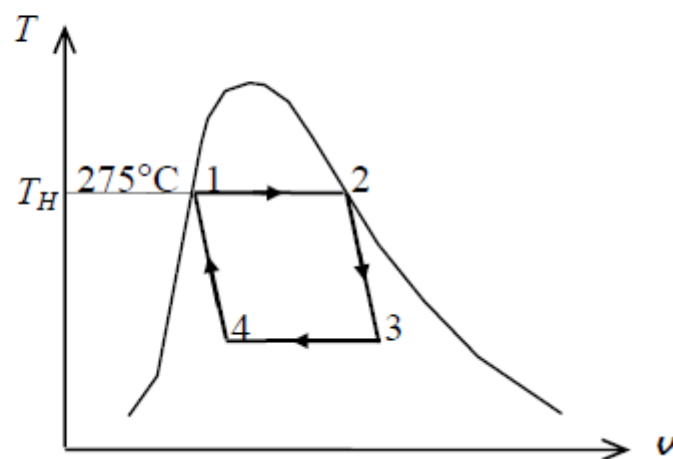
Properties The enthalpy of vaporization h_{fg} of water at 275°C is 1574.5 kJ/kg (Table A-4).

Analysis The enthalpy of vaporization h_{fg} at a given T or P represents the amount of heat transfer as 1 kg of a substance is converted from saturated liquid to saturated vapor at that T or P . Therefore, the rate of heat transfer to the steam during heat addition process is

$$\dot{Q}_H = \dot{m}h_{fg@275^{\circ}\text{C}} = (3\text{ kg/s})(1574.5\text{ kJ/kg}) = 4723\text{ kJ/s}$$

Then the power output of this heat engine becomes

$$\dot{W}_{\text{net,out}} = \eta_{\text{th}}\dot{Q}_H = (0.30)(4723\text{ kW}) = \mathbf{1417\text{ kW}}$$



6-114 A heat pump with a COP of 2.4 is used to heat a house. When running, the heat pump consumes 8 kW of electric power. If the house is losing heat to the outside at an average rate of 40,000 kJ/h and the temperature of the house is 3°C when the heat pump is turned on, determine how long it will take for the temperature in the house to rise to 22°C. Assume the house is well sealed (i.e., no air leaks) and take the entire mass within the house (air, furniture, etc.) to be equivalent to 2000 kg of air.

Analysis The house is losing heat at a rate of

$$\dot{Q}_{\text{Loss}} = 40,000 \text{ kJ/h} = 11.11 \text{ kJ/s}$$

The rate at which this heat pump supplies heat is

$$\dot{Q}_H = \text{COP}_{\text{HP}} \dot{W}_{\text{net,in}} = (2.4)(8 \text{ kW}) = 19.2 \text{ kW}$$

That is, this heat pump can supply heat at a rate of 19.2 kJ/s. Taking the house as the system (a closed system), the energy balance can be written as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{\text{system}}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc. energies}}}$$

$$Q_{\text{in}} - Q_{\text{out}} = \Delta U = m(u_2 - u_1)$$

$$Q_{\text{in}} - Q_{\text{out}} = mc_v(T_2 - T_1)$$

$$(\dot{Q}_{\text{in}} - \dot{Q}_{\text{out}})\Delta t = mc_v(T_2 - T_1)$$

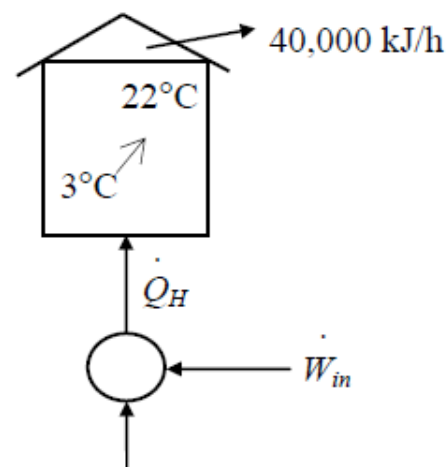
Substituting,

$$(19.2 - 11.11 \text{ kJ/s})\Delta t = (2000 \text{ kg})(0.718 \text{ kJ/kg} \cdot ^\circ\text{C})(22 - 3)^\circ\text{C}$$

Solving for Δt , it will take

$$\Delta t = 3373 \text{ s} = \mathbf{0.937 \text{ h}}$$

for the temperature in the house to rise to 22°C.



6-118 Consider a Carnot heat-engine cycle executed in a closed system using 0.01 kg of refrigerant-134a as the working fluid. The cycle has a thermal efficiency of 15 percent, and the refrigerant-134a changes from saturated liquid to saturated vapor at 50°C during the heat addition process. Determine the net work output of this engine per cycle.

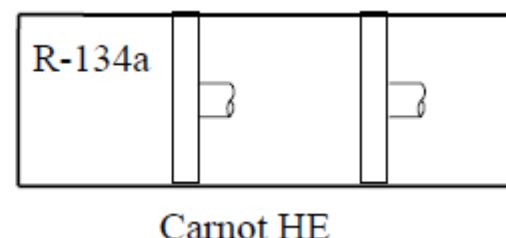
Properties The enthalpy of vaporization of R-134a at 50°C is $h_{fg} = 151.79$ kJ/kg (Table A-11).

Analysis The enthalpy of vaporization h_{fg} at a given T or P represents the amount of heat transfer as 1 kg of a substance is converted from saturated liquid to saturated vapor at that T or P . Therefore, the amount of heat transfer to R-134a during the heat addition process of the cycle is

$$Q_H = mh_{fg@50^\circ\text{C}} = (0.01 \text{ kg})(151.79 \text{ kJ/kg}) = 1.518 \text{ kJ}$$

Then the work output of this heat engine becomes

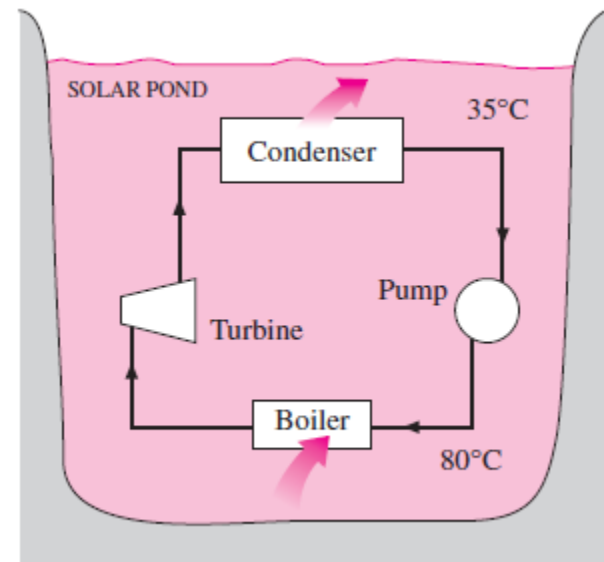
$$W_{\text{net,out}} = \eta_{\text{th}} Q_H = (0.15)(1.518 \text{ kJ}) = \mathbf{0.228 \text{ kJ}}$$



6-119 A heat pump with a COP of 2.8 is used to heat an air-tight house. When running, the heat pump consumes 5 kW of power. If the temperature in the house is 7°C when the heat pump is turned on, how long will it take for the heat pump to raise the temperature of the house to 22°C? Is this answer realistic or optimistic? Explain. Assume the entire mass within the house (air, furniture, etc.) is equivalent to 1500 kg of air. *Answer: 19.2 min*

6-120 A promising method of power generation involves collecting and storing solar energy in large artificial lakes a few meters deep, called solar ponds. Solar energy is absorbed by all parts of the pond, and the water temperature rises everywhere. The top part of the pond, however, loses to the atmosphere much of the heat it absorbs, and as a result, its temperature drops. This cool water serves as insulation for the bottom part of the pond and helps trap the energy there. Usually, salt is planted at the bottom of the pond to prevent the rise of this hot water to the top. A power plant that uses an organic fluid, such as alcohol, as the working fluid can be operated between the top and the bottom portions of the pond. If the water temperature is 35°C near the surface and 80°C near the bottom of the pond, determine the maximum thermal efficiency that this power plant can have. Is it realistic to use 35 and 80°C for temperatures in the calculations? Explain.

Answer: 12.7 percent



6-121 Consider a Carnot heat-engine cycle executed in a closed system using 0.0103 kg of steam as the working fluid. It is known that the maximum absolute temperature in the cycle is twice the minimum absolute temperature, and the net work output of the cycle is 25 kJ. If the steam changes from saturated vapor to saturated liquid during heat rejection, determine the temperature of the steam during the heat rejection process.

Assumptions The engine is said to operate on the Carnot cycle, which is totally reversible.

Analysis The thermal efficiency of the cycle is

$$\eta_{\text{th}} = 1 - \frac{T_L}{T_H} = 1 - \frac{1}{2} = 0.5$$

Also,

$$\eta_{\text{th}} = \frac{W}{Q_H} \longrightarrow Q_H = \frac{W}{\eta_{\text{th}}} = \frac{25 \text{ kJ}}{0.5} = 50 \text{ kJ}$$

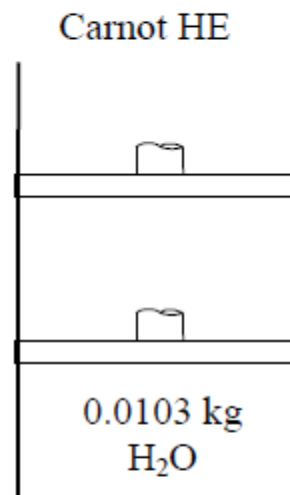
$$Q_L = Q_H - W = 50 - 25 = 25 \text{ kJ}$$

and

$$q_L = \frac{Q_L}{m} = \frac{25 \text{ kJ}}{0.0103 \text{ kg}} = 2427.2 \text{ kJ/kg} = h_{fg@T_L}$$

since the enthalpy of vaporization h_{fg} at a given T or P represents the amount of heat transfer as 1 kg of a substance is converted from saturated liquid to saturated vapor at that T or P . Therefore, T_L is the temperature that corresponds to the h_{fg} value of 2427.2 kJ/kg, and is determined from the steam tables to be

$$T_L = 31.3^\circ\text{C}$$



6-123 Consider a Carnot refrigeration cycle executed in a closed system in the saturated liquid–vapor mixture region using 0.96 kg of refrigerant-134a as the working fluid. It is known that the maximum absolute temperature in the cycle is 1.2 times the minimum absolute temperature, and the net work input to the cycle is 22 kJ. If the refrigerant changes from saturated vapor to saturated liquid during the heat rejection process, determine the minimum pressure in the cycle.

Assumptions The refrigerator is said to operate on the reversed Carnot cycle, which is totally reversible

Analysis The coefficient of performance of the cycle is

$$\text{COP}_R = \frac{1}{T_H / T_L - 1} = \frac{1}{1.2 - 1} = 5$$

Also,

$$\text{COP}_R = \frac{Q_L}{W_{\text{in}}} \longrightarrow Q_L = \text{COP}_R \times W_{\text{in}} = (5)(22 \text{ kJ}) = 110 \text{ kJ}$$

$$Q_H = Q_L + W = 110 + 22 = 132 \text{ kJ}$$

and

$$q_H = \frac{Q_H}{m} = \frac{132 \text{ kJ}}{0.96 \text{ kg}} = 137.5 \text{ kJ/kg} = h_{fg@T_H}$$

since the enthalpy of vaporization h_{fg} at a given T or P represents the amount of heat transfer per unit mass as a substance is converted from saturated liquid to saturated vapor at that T or P . Therefore, T_H is the temperature that corresponds to the h_{fg} value of 137.5 kJ/kg, and is determined from the R-134a tables to be

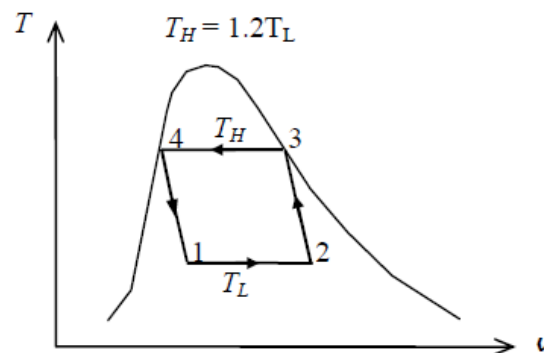
$$T_H \cong 61.3^\circ\text{C} = 334.3 \text{ K}$$

Then,

$$T_L = \frac{T_H}{1.2} = \frac{334.3 \text{ K}}{1.2} = 278.6 \text{ K} \cong 5.6^\circ\text{C}$$

Therefore,

$$P_{\text{min}} = P_{\text{sat}@5.6^\circ\text{C}} = \mathbf{355 \text{ kPa}}$$



6-125 Consider two Carnot heat engines operating in series. The first engine receives heat from the reservoir at 1800 K and rejects the waste heat to another reservoir at temperature T . The second engine receives this energy rejected by the first one, converts some of it to work, and rejects the rest to a reservoir at 300 K. If the thermal efficiencies of both engines are the same, determine the temperature T . *Answer: 735 K*

Assumptions The engines are said to operate on the Carnot cycle, which is totally reversible.

Analysis The thermal efficiency of the two Carnot heat engines can be expressed as

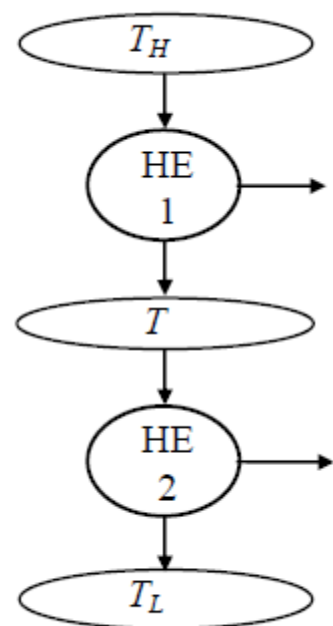
$$\eta_{\text{th,I}} = 1 - \frac{T}{T_H} \quad \text{and} \quad \eta_{\text{th,II}} = 1 - \frac{T_L}{T}$$

Equating,

$$1 - \frac{T}{T_H} = 1 - \frac{T_L}{T}$$

Solving for T ,

$$T = \sqrt{T_H T_L} = \sqrt{(1800 \text{ K})(300 \text{ K})} = \mathbf{735 \text{ K}}$$



6-128 A Carnot heat engine receives heat at 750 K and rejects the waste heat to the environment at 300 K. The entire work output of the heat engine is used to drive a Carnot refrigerator that removes heat from the cooled space at -15°C at a rate of 400 kJ/min and rejects it to the same environment at 300 K. Determine (a) the rate of heat supplied to the heat engine and (b) the total rate of heat rejection to the environment.

6-130 A heat engine operates between two reservoirs at 800 and 20°C . One-half of the work output of the heat engine is used to drive a Carnot heat pump that removes heat from the cold surroundings at 2°C and transfers it to a house maintained at 22°C . If the house is losing heat at a rate of 62,000 kJ/h, determine the minimum rate of heat supply to the heat engine required to keep the house at 22°C .

6-131 Consider a Carnot refrigeration cycle executed in a closed system in the saturated liquid–vapor mixture region using 0.8 kg of refrigerant-134a as the working fluid. The maximum and the minimum temperatures in the cycle are 20°C and -8°C , respectively. It is known that the refrigerant is saturated liquid at the end of the heat rejection process, and the net work input to the cycle is 15 kJ. Determine the fraction of the mass of the refrigerant that vaporizes during the heat addition process, and the pressure at the end of the heat rejection process.

6-132 Consider a Carnot heat-pump cycle executed in a steady-flow system in the saturated liquid–vapor mixture region using refrigerant-134a flowing at a rate of 0.264 kg/s as the working fluid. It is known that the maximum absolute temperature in the cycle is 1.25 times the minimum absolute temperature, and the net power input to the cycle is 7 kW. If the refrigerant changes from saturated vapor to saturated liquid during the heat rejection process, determine the ratio of the maximum to minimum pressures in the cycle.

6-133 A Carnot heat engine is operating between a source at T_H and a sink at T_L . If it is desired to double the thermal efficiency of this engine, what should the new source temperature be? Assume the sink temperature is held constant.

Analysis Denoting the new source temperature by T_H^* , the thermal efficiency of the Carnot heat engine for both cases can be expressed as

$$\eta_{\text{th,C}} = 1 - \frac{T_L}{T_H} \quad \text{and} \quad \eta_{\text{th,C}}^* = 1 - \frac{T_L}{T_H^*} = 2\eta_{\text{th,C}}$$

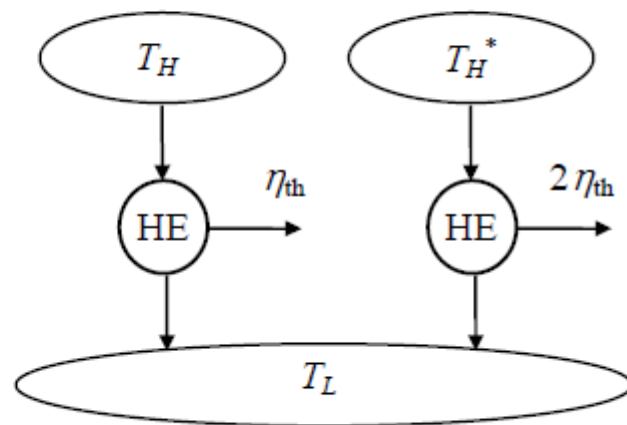
Substituting,

$$1 - \frac{T_L}{T_H^*} = 2 \left(1 - \frac{T_L}{T_H} \right)$$

Solving for T_H^* ,

$$T_H^* = \frac{T_H T_L}{T_H - 2T_L}$$

which is the desired relation.



6-134 When discussing Carnot engines, it is assumed that the engine is in thermal equilibrium with the source and the sink during the heat addition and heat rejection processes, respectively. That is, it is assumed that $T_H^* = T_H$ and $T_L^* = T_L$ so that there is no external irreversibility. In that case, the thermal efficiency of the Carnot engine is $\eta_C = 1 - T_L/T_H$.

In reality, however, we must maintain a reasonable temperature difference between the two heat transfer media in order to have an acceptable heat transfer rate through a finite heat exchanger surface area. The heat transfer rates in that case can be expressed as

$$\dot{Q}_H = (hA)_H(T_H - T_H^*)$$

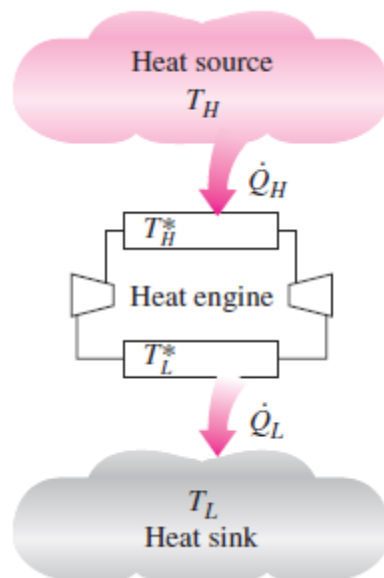
$$\dot{Q}_L = (hA)_L(T_L^* - T_L)$$

where h and A are the heat transfer coefficient and heat transfer surface area, respectively. When the values of h , A , T_H , and T_L are fixed, show that the power output will be a maximum when

$$\frac{T_L^*}{T_H} = \left(\frac{T_L}{T_H} \right)^{1/2}$$

Also, show that the maximum net power output in this case is

$$\dot{W}_{C,\max} = \frac{(hA)_H T_H}{1 + (hA)_H / (hA)_L} \left[1 - \left(\frac{T_L}{T_H} \right)^{1/2} \right]^2$$



Analysis It is given that

$$\dot{Q}_H = (hA)_H (T_H - T_H^*).$$

Therefore,

$$\dot{W} = \eta_{th} \dot{Q}_H = \left(1 - \frac{T_L^*}{T_H^*}\right) (hA)_H (T_H - T_H^*) = \left(1 - \frac{T_L^*}{T_H^*}\right) (hA)_H \left(1 - \frac{T_H^*}{T_H}\right) T_H$$

or,

$$\frac{\dot{W}}{(hA)_H T_H} = \left(1 - \frac{T_L^*}{T_H^*}\right) \left(1 - \frac{T_H^*}{T_H}\right) = (1-r)x \quad (1)$$

where we defined r and x as $r = T_L^*/T_H^*$ and $x = 1 - T_H^*/T_H$.

For a reversible cycle we also have

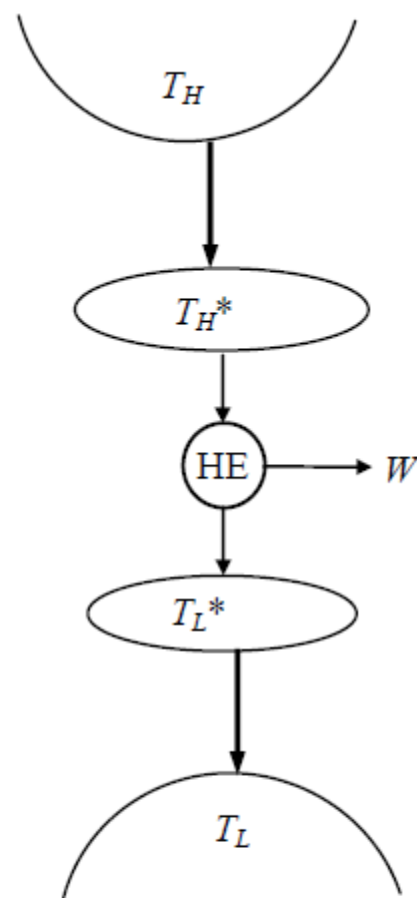
$$\frac{T_H^*}{T_L^*} = \frac{\dot{Q}_H}{\dot{Q}_L} \longrightarrow \frac{1}{r} = \frac{(hA)_H (T_H - T_H^*)}{(hA)_L (T_L^* - T_L)} = \frac{(hA)_H T_H (1 - T_H^*/T_H)}{(hA)_L T_H (T_L^*/T_H - T_L/T_H)}$$

but

$$\frac{T_L^*}{T_H} = \frac{T_L^*}{T_H^*} \frac{T_H^*}{T_H} = r(1-x).$$

Substituting into above relation yields

$$\frac{1}{r} = \frac{(hA)_H x}{(hA)_L [r(1-x) - T_L/T_H]}$$



Solving for x,

$$x = \frac{r - T_L / T_H}{r[(hA)_H / (hA)_L + 1]} \quad (2)$$

Substitute (2) into (1):

$$\dot{W} = (hA)_H T_H (1 - r) \frac{r - T_L / T_H}{r[(hA)_H / (hA)_L + 1]} \quad (3)$$

Taking the partial derivative $\frac{\partial \dot{W}}{\partial r}$ holding everything else constant and setting it equal to zero gives

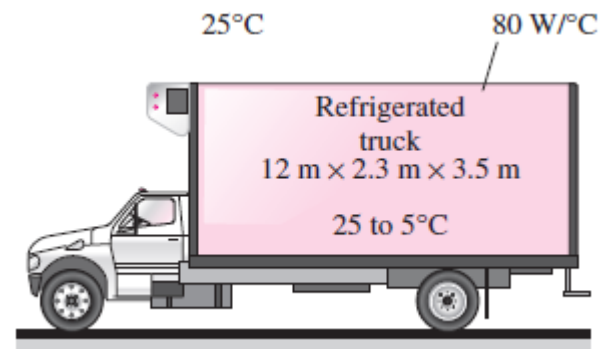
$$r = \frac{T_L^*}{T_H^*} = \left(\frac{T_L}{T_H} \right)^{1/2} \quad (4)$$

which is the desired relation. The maximum net power output in this case is determined by substituting (4) into (3). It simplifies to

$$\dot{W}_{\max} = \frac{(hA)_H T_H}{1 + (hA)_H / (hA)_L} \left\{ 1 - \left(\frac{T_L}{T_H} \right)^{1/2} \right\}^2$$

6-135 Replacing incandescent lights with energy-efficient fluorescent lights can reduce the lighting energy consumption to one-fourth of what it was before. The energy consumed by the lamps is eventually converted to heat, and thus switching to energy-efficient lighting also reduces the cooling load in summer but increases the heating load in winter. Consider a building that is heated by a natural gas furnace with an efficiency of 80 percent and cooled by an air conditioner with a COP of 3.5. If electricity costs \$0.08/kWh and natural gas costs \$1.40/therm, determine if efficient lighting will increase or decrease the total energy cost of the building (*a*) in summer and (*b*) in winter.

6-136 The cargo space of a refrigerated truck whose inner dimensions are $12\text{ m} \times 2.3\text{ m} \times 3.5\text{ m}$ is to be precooled from 25°C to an average temperature of 5°C . The construction of the truck is such that a transmission heat gain occurs at a rate of $80\text{ W}/^\circ\text{C}$. If the ambient temperature is 25°C , determine how long it will take for a system with a refrigeration capacity of 8 kW to precool this truck.



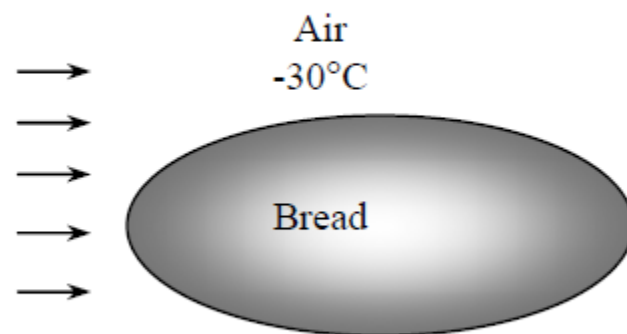
6-137 A refrigeration system is to cool bread loaves with an average mass of 450 g from 22 to -10°C at a rate of 500 loaves per hour by refrigerated air at -30°C . Taking the average specific and latent heats of bread to be $2.93 \text{ kJ/kg} \cdot ^{\circ}\text{C}$ and 109.3 kJ/kg , respectively, determine (a) the rate of heat removal from the breads, in kJ/h ; (b) the required volume flow rate of air, in m^3/h , if the temperature rise of air is not to exceed 8°C ; and (c) the size of the compressor of the refrigeration system, in kW , for a COP of 1.2 for the refrigeration system.

Assumptions 1 Steady operating conditions exist. 2 The thermal properties of the bread loaves are constant. 3 The cooling section is well-insulated so that heat gain through its walls is negligible.

Properties The average specific and latent heats of bread are given to be $2.93 \text{ kJ/kg} \cdot ^{\circ}\text{C}$ and 109.3 kJ/kg , respectively. The gas constant of air is $0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}$ (Table A-1), and the specific heat of air at the average temperature of $(-30 + -22)/2 = -26^{\circ}\text{C} \approx 250 \text{ K}$ is $c_p = 1.0 \text{ kJ/kg} \cdot ^{\circ}\text{C}$ (Table A-2).

Analysis (a) Noting that the breads are cooled at a rate of 500 loaves per hour, breads can be considered to flow steadily through the cooling section at a mass flow rate of

$$\dot{m}_{\text{bread}} = (500 \text{ breads/h})(0.45 \text{ kg/bread}) = 225 \text{ kg/h} = 0.0625 \text{ kg/s}$$



Then the rate of heat removal from the breads as they are cooled from 22°C to -10°C and frozen becomes

$$\begin{aligned}\dot{Q}_{\text{bread}} &= (\dot{m} c_p \Delta T)_{\text{bread}} = (225 \text{ kg/h})(2.93 \text{ kJ/kg} \cdot ^\circ\text{C})[(22 - (-10))^\circ\text{C}] \\ &= 21,096 \text{ kJ/h}\end{aligned}$$

$$\dot{Q}_{\text{freezing}} = (\dot{m} h_{\text{latent}})_{\text{bread}} = (225 \text{ kg/h})(109.3 \text{ kJ/kg}) = 24,593 \text{ kJ/h}$$

and

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{bread}} + \dot{Q}_{\text{freezing}} = 21,096 + 24,593 = \mathbf{45,689 \text{ kJ/h}}$$

(b) All the heat released by the breads is absorbed by the refrigerated air, and the temperature rise of air is not to exceed 8°C. The minimum mass flow and volume flow rates of air are determined to be

$$\dot{m}_{\text{air}} = \frac{\dot{Q}_{\text{air}}}{(c_p \Delta T)_{\text{air}}} = \frac{45,689 \text{ kJ/h}}{(1.0 \text{ kJ/kg} \cdot ^\circ\text{C})(8^\circ\text{C})} = 5711 \text{ kg/h}$$

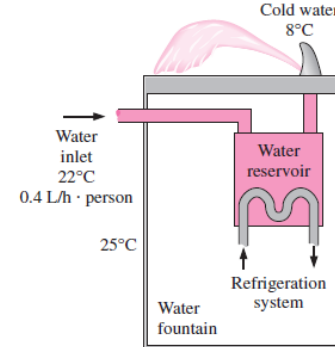
$$\rho = \frac{P}{RT} = \frac{101.3 \text{ kPa}}{(0.287 \text{ kPa} \cdot \text{m}^3 / \text{kg} \cdot \text{K})(-30 + 273) \text{ K}} = 1.45 \text{ kg/m}^3$$

$$\dot{V}_{\text{air}} = \frac{\dot{m}_{\text{air}}}{\rho_{\text{air}}} = \frac{5711 \text{ kg/h}}{1.45 \text{ kg/m}^3} = \mathbf{3939 \text{ m}^3/\text{h}}$$

(c) For a COP of 1.2, the size of the compressor of the refrigeration system must be

$$\dot{W}_{\text{refrig}} = \frac{\dot{Q}_{\text{refrig}}}{\text{COP}} = \frac{45,689 \text{ kJ/h}}{1.2} = 38,074 \text{ kJ/h} = \mathbf{10.6 \text{ kW}}$$

6-138 The drinking water needs of a production facility with 20 employees is to be met by a bubbler type water fountain. The refrigerated water fountain is to cool water from 22 to 8°C and supply cold water at a rate of 0.4 L per hour per person. Heat is transferred to the reservoir from the surroundings at 25°C at a rate of 45 W. If the COP of the refrigeration system is 2.9, determine the size of the compressor, in W, that will be suitable for the refrigeration system of this water cooler.



Analysis The refrigeration load in this case consists of the heat gain of the reservoir and the cooling of the incoming water. The water fountain must be able to provide water at a rate of

$$\dot{m}_{\text{water}} = \rho \dot{V}_{\text{water}} = (1 \text{ kg/L})(0.4 \text{ L/h} \cdot \text{person})(20 \text{ persons}) = 8.0 \text{ kg/h}$$

To cool this water from 22°C to 8°C, heat must be removed from the water at a rate of

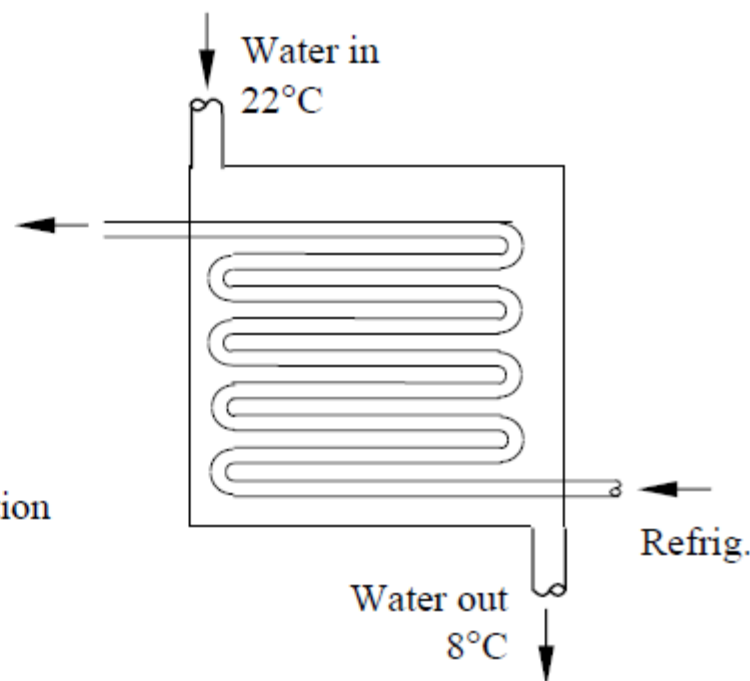
$$\begin{aligned} \dot{Q}_{\text{cooling}} &= \dot{m} c_p (T_{\text{in}} - T_{\text{out}}) \\ &= (8.0 \text{ kg/h})(4.18 \text{ kJ/kg} \cdot ^\circ\text{C})(22 - 8)^\circ\text{C} \\ &= 468 \text{ kJ/h} = 130 \text{ W} \quad (\text{since } 1 \text{ W} = 3.6 \text{ kJ/h}) \end{aligned}$$

Then total refrigeration load becomes

$$\dot{Q}_{\text{refrig, total}} = \dot{Q}_{\text{cooling}} + \dot{Q}_{\text{transfer}} = 130 + 45 = 175 \text{ W}$$

Noting that the coefficient of performance of the refrigeration system is 2.9, the required power input is

$$\dot{W}_{\text{refrig}} = \frac{\dot{Q}_{\text{refrig}}}{\text{COP}} = \frac{175 \text{ W}}{2.9} = \mathbf{60.3 \text{ W}}$$

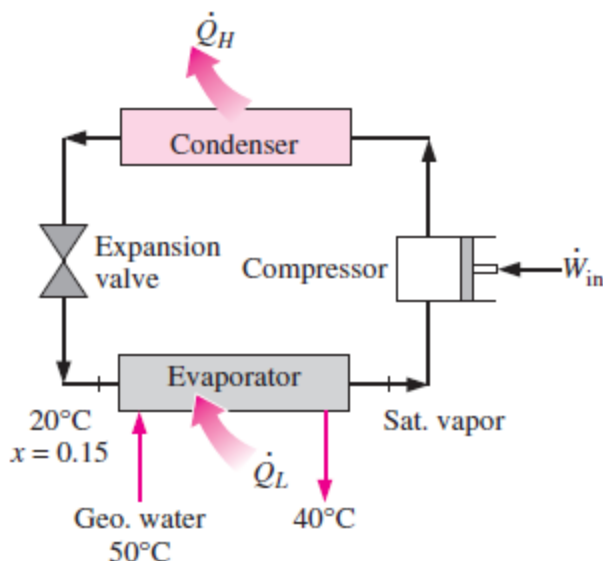


Therefore, the power rating of the compressor of this refrigeration system must be at least 60.3 W to meet the cold water requirements of this office.

6-145 The kitchen, bath, and other ventilation fans in a house should be used sparingly since these fans can discharge a houseful of warmed or cooled air in just one hour. Consider a 200-m² house whose ceiling height is 2.8 m. The house is heated by a 96 percent efficient gas heater and is maintained at 22°C and 92 kPa. If the unit cost of natural gas is \$1.20/therm (1 therm = 105,500 kJ), determine the cost of energy “vented out” by the fans in 1 h. Assume the average outdoor temperature during the heating season to be 5°C.

6-146 Repeat Prob. 6-145 for the air-conditioning cost in a dry climate for an outdoor temperature of 28°C. Assume the COP of the air-conditioning system to be 2.3, and the unit cost of electricity to be \$0.10/kWh.

6-148 A heat pump with refrigerant-134a as the working fluid is used to keep a space at 25°C by absorbing heat from geothermal water that enters the evaporator at 50°C at a rate of 0.065 kg/s and leaves at 40°C. Refrigerant enters the evaporator at 20°C with a quality of 15 percent and leaves at the same pressure as saturated vapor. If the compressor consumes 1.2 kW of power, determine (a) the mass flow rate of the refrigerant, (b) the rate of heat supply, (c) the COP, and (d) the minimum power input to the compressor for the same rate of heat supply. *Answers: (a) 0.0175 kg/s, (b) 3.92 kW, (c) 3.27, (d) 0.303 kW*



6-149 Cold water at 10°C enters a water heater at the rate of $0.02 \text{ m}^3/\text{min}$ and leaves the water heater at 50°C . The water heater receives heat from a heat pump that receives heat from a heat source at 0°C .

(a) Assuming the water to be an incompressible liquid that does not change phase during heat addition, determine the rate of heat supplied to the water, in kJ/s .

(b) Assuming the water heater acts as a heat sink having an average temperature of 30°C , determine the minimum power supplied to the heat pump, in kW .

Properties The specific heat and specific volume of water at room temperature are $c_p = 4.18 \text{ kJ/kg}\cdot\text{K}$ and $\nu = 0.001 \text{ m}^3/\text{kg}$ (Table A-3).

Analysis (a) An energy balance on the water heater gives the rate of heat supplied to the water

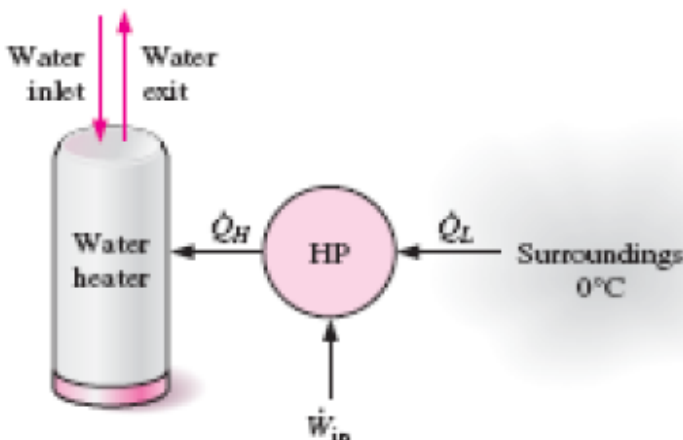
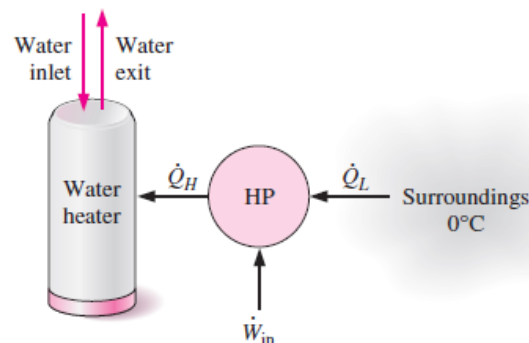
$$\begin{aligned}\dot{Q}_H &= \dot{m}c_p(T_2 - T_1) \\ &= \frac{\dot{V}}{\nu}c_p(T_2 - T_1) \\ &= \frac{(0.02/60) \text{ m}^3/\text{s}}{0.001 \text{ m}^3/\text{kg}}(4.18 \text{ kJ/kg}\cdot^\circ\text{C})(50 - 10)^\circ\text{C} \\ &= \mathbf{55.73 \text{ kW}}\end{aligned}$$

(b) The COP of a reversible heat pump operating between the specified temperature limits is

$$\text{COP}_{\max} = \frac{1}{1 - T_L / T_H} = \frac{1}{1 - (0 + 273)/(30 + 273)} = 10.1$$

Then, the minimum power input would be

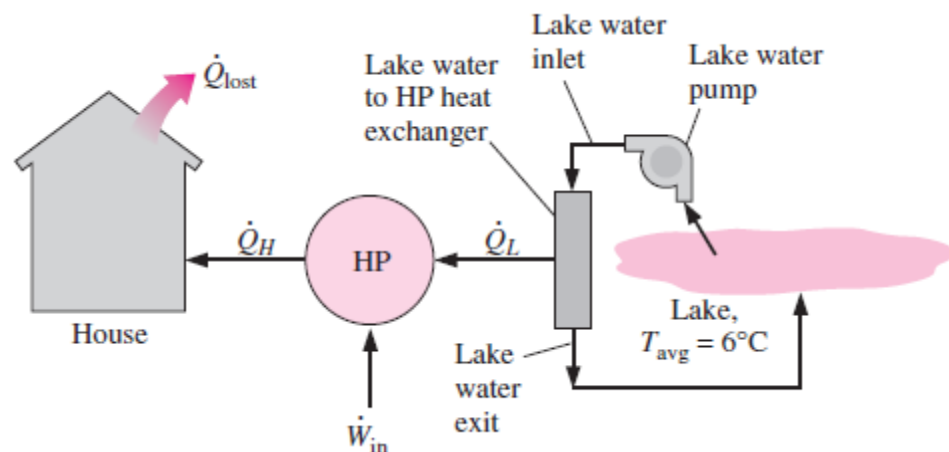
$$\dot{W}_{\text{in},\min} = \frac{\dot{Q}_H}{\text{COP}_{\max}} = \frac{55.73 \text{ kW}}{10.1} = \mathbf{5.52 \text{ kW}}$$



6-150 A heat pump receives heat from a lake that has an average winter time temperature of 6°C and supplies heat into a house having an average temperature of 27°C .

(a) If the house loses heat to the atmosphere at the rate of $64,000\text{ kJ/h}$, determine the minimum power supplied to the heat pump, in kW.

(b) A heat exchanger is used to transfer the energy from the lake water to the heat pump. If the lake water temperature decreases by 5°C as it flows through the lake water-to-heat pump heat exchanger, determine the minimum mass flow rate of lake water, in kg/s. Neglect the effect of the lake water pump.



6-151 A heat pump supplies heat energy to a house at the rate of $140,000\text{ kJ/h}$ when the house is maintained at 25°C . Over a period of one month, the heat pump operates for 100 hours to transfer energy from a heat source outside the house to inside the house. Consider a heat pump receiving heat from two different outside energy sources. In one application the heat pump receives heat from the outside air at 0°C . In a second application the heat pump receives heat from a lake having a water temperature of 10°C . If electricity costs $\$0.085/\text{kWh}$, determine the maximum money saved by using the lake water rather than the outside air as the outside energy source.

Chapter 6

End of the Problems